

1 Impermanent loss and slippage in Automated Market Makers
2 (AMMs) with constant-product formula

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7 **Abstract**

8 Automated Market Makers (AMMs) are the main liquidity providers in the Decentralised
9 Finance (DeFi) space. They are programmes that follow some coded rules to determine and
10 adjust prices and volumes. Amongst those rules, the most common one is the constant-product
11 formula, which is followed by the most important AMMs in DeFi: Uniswap in the Ethereum
12 blockchain [Unia] and Raydium in the Solana blockchain [Raya]. In this paper we explain the
13 mathematical formulas that are used to compute prices and volumes in AMMs with constant-
14 product formulas, and utilise this framework to study the effect of price and volume changes on
15 AMMs from two points of view: (1) the liquidity provider's (i.e. the AMM) and (2) the trader's
16 (who initiates the swap). For the liquidity provider we analyse the impermanent loss and how
17 it behaves with respect to changes in price and available liquidity. For the trader we compute
18 the slippage and show that this is exactly the impermanent gain (i.e. negative impermanent
19 loss) of the AMM. We also analyse the effect of liquidity balance/imbalance in the AMM on the
20 resulting swap prices: the AMM will favour swaps that help it to maintain a balanced pool and
21 penalise swaps that push the pool out of its balance. Finally, we show how the “impermanent
22 loss” behaves with respect to reversion in prices: the loss will disappear if there is a complete
23 u-turn in prices (hence the adjective “impermanent”), but if we perform some actions before
24 the price reversion is complete the impermanent loss can become permanent.

25 **Contents**

26	1 Automated Market Makers (AMMs) and traditional market makers (MMs)	2
27	2 Mathematical framework for AMMs	3
28	2.1 Rules and equations governing the AMM mechanics	3
29	2.2 Deducing the Balanced liquidity provision	4
30	2.3 Duality between prices and token amounts	6

31	3	Impermanent loss and price changes	6
32	3.1	Token X rises to 6 USD	7
33	3.2	Token X drops to 4 USD	9
34	3.3	Under a price change, the impermanent loss is invariant with respect to the initial	
35		liquidity	10
36	4	Impermanent loss and liquidity consumption	11
37	4.1	Trader buys 5 tokens X from the AMM	11
38	4.2	Trader sells 5 tokens X to the AMM	12
39	4.3	Impermanent gain and slippage are decreasing in the available liquidity	13
40	5	Slippage and liquidity balance/imbalance within the AMM	13
41	5.1	Trader buys in a balanced liquidity pool	14
42	5.2	Trader buys the token with smaller liquidity in the pool	15
43	5.3	Trader buys the token with greater liquidity in the pool	15
44	6	Impermanent loss and price reversion	16
45	6.1	The impermanent loss disappears when there is price reversion	16
46	6.2	The impermanent loss may become permanent if some liquidity is withdrawn	18
47	7	Conclusions	19
48	8	See also	20

49 **1 Automated Market Makers (AMMs) and traditional mar-** 50 **ket makers (MMs)**

51 Automated Market Makers (AMMs) play the role of liquidity providers in protocols or projects in
52 the Decentralised Finance (DeFi) space. AMMs are programmes, hence they operate without human
53 intervention, following a number of rules that govern the way these AMMs provide liquidity and
54 adjust prices.

55 A traditional market maker (MM) buys and sells an asset, providing liquidity to potential buyers
56 or sellers. The MM will quote two prices: the ask price at which they are willing to sell, and the
57 bid price at which they are willing to buy. The difference between these two prices is known as the
58 (bid-ask) spread; this spread is the remuneration of the MM for providing liquidity: they sell at a
59 higher price and buy at a lower price.

60 When you trade with an automated Market Maker (AMM), instead of buying and selling one
61 token you enter a swap i.e. you exchange one token for another token. This sounds different than
62 the MM, but it is not really: when you buy an asset to an MM you pay them with a currency, say
63 USD; when you sell the asset you provide the latter to the MM in exchange of a certain amount of
64 USD. Therefore, if we see the asset as token X and the USD currency as token Y then MMs and
65 AMM are actually the same.

66 The key concept of spread for an MM is translated as slippage cost for AMMs. Indeed, just like
67 the spread helps the MM to avoid adverse selection (in a nutshell, to be on the wrong side of the
68 trade when prices have a trend), the slippage helps the AMM to keep in balance the two assets in
69 its liquidity pool. More precisely, the slippage in the AMM can be thought as the half spread in
70 the MM: in both cases these two quantities are the extra cost relative to the market price (current
71 swap value for the AMM, current mid price for the MM) that the trader incurs when dealing with
72 an AMM or a MM.

73 2 Mathematical framework for AMMs

74 2.1 Rules and equations governing the AMM mechanics

75 There are a few rules for AMMs to compute the right prices for a swap and calculate the corre-
76 sponding impermanent loss for the trade. Let us assume that we have two tokens in a pool, namely
77 token X and token Y , and that we are the only liquidity providers for the pool. For any time
78 $t \geq 0$ we denote $x(t)$ the total number of tokens X in the pool, $P_X(t)$ the price of token X and
79 $N_X := x(t)P_X(t)$ the notional of token X i.e. its value in USD. The variables corresponding to token
80 Y are $y(t)$ for the number of tokens Y , $P_Y(t)$ for their price and $N_Y(t)$ for their notional.

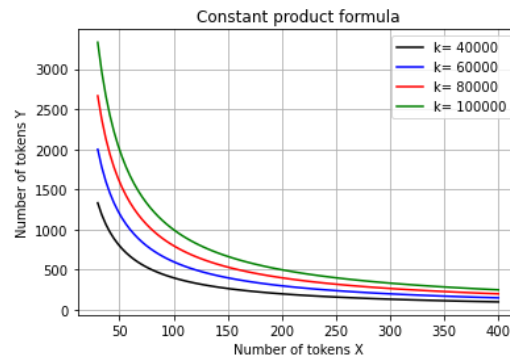


Figure 1: Constant-product formula for different values of k . Notice that the price adjustment is a power law, which is superlinear.

81 1. **Constant-product formula.** The available liquidity in tokens within the AMM is invariant:

$$x(t)y(t) = k \quad \text{for all } t \geq 0 \quad (1)$$

82 where $k > 0$ is constant (see Figure 1).

83 2. **Balanced liquidity provision.** We start with a balanced liquidity provision, i.e. we deposit
84 a 50/50 distribution of USD value of token X and Y into the AMM. This implies that our
85 initial notionals are equal

$$N_X(0) = N_Y(0)$$

86 No matter what happens during the time interval $[0, T]$, our liquidity provision should always
 87 remain at equilibrium i.e.

$$N_X(t) = x(t)P_X(t) = y(t)P_Y(t) = N_Y(t) \quad \text{for all } t \in [0, T]. \quad (2)$$

88 It is important to remark that Equation (2) is not an extra assumption but rather a consequence
 89 of choosing the product-invariant formula (1). We will prove this result in Subsection 2.2 below.

90 **3. Liquidity position.** Our total liquidity provision in USD within the AMM pool is:

$$Liquidity(t) = N_X(t) + N_Y(t) = x(t)P_X(t) + y(t)P_Y(t) \quad \text{for all } t \geq 0. \quad (3)$$

91 **4. Hold position.** The value of the position if we held the tokens X and Y instead of using
 92 them to provide liquidity to the AMM:

$$Hold(t) = x(0)P_X(t) + y(0)P_Y(t) \quad \text{for all } t \geq 0. \quad (4)$$

93 Observe that both the liquidity and hold positions are marked to market i.e. prices are com-
 94 puted at the current time t . However, the number of tokens in the hold position is constant
 95 and equal to the initial liquidity provision.

96 **5. Impermanent loss.** It is the difference between the hold position and the liquidity position,
 97 either in USD or in percentage points

$$\begin{aligned} LossUsd(t) &= Hold(t) - Liquidity(t), \\ LossPct(t) &= \frac{LossUsd(t)}{Hold(t)}, \end{aligned} \quad (5)$$

98 for all $t \geq 0$.

99 2.2 Deducing the Balanced liquidity provision

100 Suppose that we have a function $f(x, y)$ and that we can only choose points (x, y) that satisfy a
 101 constraint of the form $g(x, y) = k$. If the point (x_0, y_0) is a maximum or minimum of $f(x, y)$ along
 102 the curve $g(x, y) = k$ then there exists $\lambda \neq 0$ such that

$$\nabla f(x_0, y_0) = \lambda \nabla g(x_0, y_0).$$

103 Moreover, critical points of $f(x, y)$ under the constraint $g(x, y) = k$ are critical points of the uncon-
 104 strained functional (called *Lagrangian*)

$$L(x, y, \lambda) = f(x, y) - \lambda[g(x, y) - k]. \quad (6)$$

105 In other words, with Equation (6) we transformed the 2-dimensional problem of maximising $f(x, y)$
 106 under the constraint $g(x, y) = k$ to the 3-dimensional problem of maximising $L(x, y, \lambda)$ without
 107 constraints. For a proof of this result, which is called the “Lagrange Multiplier Theorem”, we
 108 recommend to check [MT11] and/or [Wik].

109 Let us come back to our case of an AMM. The constraint is the constant-product formula i.e.

$$g(x, y) = xy.$$

110 The function f that we aim to maximise is the total liquidity provision (3) i.e.

$$f(x, y) = xP_X + yP_Y. \quad (7)$$

111 By solving this problem, what we do is to find the optimal number of tokens (x, y) that maximise
 112 the USD value of the total liquidity provision, under the constraint of the constant-product formula.

113 If (x_0, y_0, λ_0) is a critical point of the Lagrangian

$$L(x, y, \lambda) = xP_X + yP_Y - \lambda[xy - k]$$

114 then the following equations are satisfied:

$$\begin{aligned} \frac{\partial L}{\partial x}(x_0, y_0, \lambda_0) &= P_X - \lambda_0 y_0 = 0, \\ \frac{\partial L}{\partial y}(x_0, y_0, \lambda_0) &= P_Y - \lambda_0 x_0 = 0, \\ \frac{\partial L}{\partial \lambda}(x_0, y_0, \lambda_0) &= x_0 y_0 - k = 0. \end{aligned} \quad (8)$$

115 The last equation in (8) simply states that the critical point (x_0, y_0) satisfies the constraint i.e. the
 116 constant-product formula $xy = k$. From the first and second equations in (8) we obtain

$$\frac{P_X}{y_0} = \lambda_0 = \frac{P_Y}{x_0},$$

117 which is exactly (2), as we claimed.

118
 119 The principles we applied here are quite standard in Microeconomics. Indeed, the function (7)
 120 is the Budget Line, the product xy is the Utility Function, and for any $k > 0$ the constant-product
 121 curve $xy = k$ is called an Indifference Curve: the AMM is indifferent of the choice of a point on the
 122 curve since all of them have the same utility, namely k . Therefore, what we did in this section was
 123 to maximise the budget under the constraint of a fixed utility level k , given the current prices P_X
 124 and P_Y . For more details on Budget Lines and Indifference curves we suggest to start with [Hel].

2.3 Duality between prices and token amounts

There is a “principle of duality” in Equation (2), in the sense that we can either assume that the prices $P_X(t)$ and $P_Y(t)$ are given and then compute the token amounts $x(t)$ and $y(t)$, or either assume that the token amounts are given and compute the implied prices that satisfy the balance condition (2). Let us elaborate this point in detail.

On the one hand we have a traditional MM. As we assumed in Subsection 2.2, the MM will have the two prices $P_X(t)$ and $P_Y(t)$ as exogenous or independent variables, and then they will optimally adjust their liquidity provision $x(t)$ and $y(t)$ to satisfy the two constraints, namely the constant-product formula (1) and the constant-notional formula or balanced liquidity provision (2).

On the other hand, we have an AMM like Uniswap in the Ethereum blockchain [Unia]. If the prices $P_X(t)$ and $P_Y(t)$ were coming from an external source then data risk on the price feed that is used, but also a code risk on the actual program within the AMM (known as “smart contract”) that computes the optimal liquidity distribution $x(t)$ and $y(t)$. In order to eliminate these two sources of risk, Uniswap uses what we call the “principle of duality”: instead of having the prices $P_X(t)$ and $P_Y(t)$ as independent variables and the token amounts $x(t)$ and $y(t)$ as dependent variables (like the traditional MM does), Uniswap (or any other AMM) uses the token amounts as independent variables and compute the prices endogenously as a function of these token amounts.

In the light of this “principle of duality” between prices $P_X(t)$ and $P_Y(t)$ and token amounts $x(t)$ and $y(t)$, we can either consider (1) that the prices move first (based on the chosen price feed) and then the token amounts have to be optimally adjusted by the MM, or (2) that the token amounts move first (due to swaps arriving at the AMM or changes in its liquidity provision) and then the endogenous prices are optimally adjusted. Regardless of whether we take the point of view of the MM or the AMM, in both cases the end result is that the balance condition (2) holds.

In our subsequent analysis and examples, we will assume the point of view of a MM: just like ourselves, we assume that the reader is familiar with basic economic and financial concepts like utility functions, budget lines, market making and liquidity provision. Therefore, we will assume that price changes occur first and then and compute the corresponding token amounts. However, the reader can take the point of view of the AMM without any loss of generality: in this framework, the formulas we will present are reverse-engineering the token amounts (which occurred first) from their impact in the observable prices (which occurred second).

3 Impermanent loss and price changes

Let us start with the following values:

$$\begin{aligned}P_X(0) &= 5 \text{ USD}, \\P_Y(0) &= 2 \text{ USD}.\end{aligned}$$

Let us assume that, at start, the AMM is empty and we are the only liquidity providers. In order to comply with Rule (2) i.e. a balanced liquidity provision (equilibrium condition) we need to

159 provide the same liquidity in USD for both tokens. For this reason, we will assume

$$\begin{aligned} N_X(0) &= 500 \text{ USD}, \\ N_Y(0) &= 500 \text{ USD}. \end{aligned}$$

160 Therefore, the initial liquidity provision in tokens is

$$\begin{aligned} x(0) &= 100 \text{ tokens}, \\ y(0) &= 250 \text{ tokens}. \end{aligned}$$

161 From these values we can readily compute the constant in Rule (1) (constant-product formula):

$$k = x(0)y(0) = 100 \times 250 = 25,000.$$

162 **3.1 Token X rises to 6 USD**

163 Suppose that at time $t > 0$ there is a change in prices:

$$\begin{aligned} P_X(t) &= 6 \text{ USD}, \\ P_Y(t) &= 2 \text{ USD}. \end{aligned}$$

164 Since the equilibrium condition (2) must hold then we have

$$y(t) = \frac{P_X(t)}{P_Y(t)} x(t). \quad (9)$$

165 From Equations (1) and (9) we obtain

$$k = x(t)y(t) = x(t) \times \frac{P_X(t)}{P_Y(t)} x(t) = \frac{P_X(t)}{P_Y(t)} x^2(t).$$

166 In consequence,

$$x(t) = \sqrt{k \frac{P_Y(t)}{P_X(t)}}. \quad (10)$$

167 Equations (9) and (10) are everything we need to re-calibrate the number of tokens X and Y
 168 whenever a price change occurs. After some straightforward algebra we can make $x(t)$ and $y(t)$ to
 169 be dependent only on the prices $P_X(t)$ and $P_Y(t)$. In the end we arrive to a neat expression for $x(t)$
 170 and $y(t)$, that we place below and inside a nice box for posterity:

$$\boxed{\begin{aligned} x(t) &= \sqrt{k \frac{P_Y(t)}{P_X(t)}}, \\ y(t) &= \sqrt{k \frac{P_X(t)}{P_Y(t)}}. \end{aligned}} \quad (11)$$

171 Let us now compute all quantities at time t :

$$\begin{aligned} x(t) &= \sqrt{k \frac{P_Y(t)}{P_X(t)}} \\ &= \sqrt{25,000 \times \frac{2}{6}} \\ &= 91.2871 \text{ tokens } X, \end{aligned}$$

172

$$\begin{aligned} y(t) &= \sqrt{k \frac{P_X(t)}{P_Y(t)}} \\ &= \sqrt{25,000 \times \frac{6}{2}} \\ &= 273.8613 \text{ tokens } Y, \end{aligned}$$

173

$$\begin{aligned} Liquidity(t) &= x(t)P_X(t) + y(t)P_Y(t) \\ &= 91.2871 \times 6 + 273.8613 \times 2 \\ &= 1,095.4452 \text{ USD}, \end{aligned}$$

174

$$\begin{aligned} Hold(t) &= x(0)P_X(t) + y(0)P_Y(t) \\ &= 100 \times 6 + 250 \times 2 \\ &= 1,100 \text{ USD}, \end{aligned}$$

175

$$\begin{aligned} LossUsd(t) &= Hold(t) - Liquidity(t) \\ &= 1,100 - 1,095.4452 \\ &= 4.5548 \text{ USD}, \end{aligned}$$

176

$$\begin{aligned} LossPct(t) &= \frac{LossUsd(t)}{Hold(t)} \\ &= \frac{4.5548}{1,100} \\ &= 0.414\%. \end{aligned}$$

177 Observe than in this case both the hold and the liquidity positions are profitable: both notionals
178 are above the original 1,000 USD (500 USD for each token X and Y). However, there is an imperma-
179 nent loss of 0.414%: this means that by providing liquidity to the AMM we obtain a slightly smaller
180 gain than if we just held the tokens: the total notionals are 1,095.4452 and 1,100 USD respectively.

181 We should not mistake our liquidity provision with the total liquidity available in the AMM.
182 Indeed, in the example above we started depositing 100 tokens X and 250 tokens Y into the AMM,

183 and in exchange we received some Liquidity Provision (LP) tokens issued by the AMM owner. After
 184 the price move, if we redeem all of our LP tokens we will receive 91.2871 tokens X and 273.8613
 185 tokens Y . The total value of the LP token is thus 1,095.4452 USD, which includes a small loss
 186 relative to holding the tokens instead of providing liquidity. However, this loss will only be realised
 187 if we redeem the LP tokens: if the price of token X reverts back to 5 USD then the loss disappears:
 188 this is why it is called “impermanent loss”. We will discuss this feature in great detail later on.

189 3.2 Token X drops to 4 USD

190 The initial conditions are the same as before:

$$\begin{aligned} P_X(0) &= 5 \text{ USD}, \\ P_Y(0) &= 2 \text{ USD}, \\ N_X(0) &= 500 \text{ USD}, \\ N_Y(0) &= 500 \text{ USD}, \\ x(0) &= 100 \text{ tokens } X, \\ y(0) &= 250 \text{ tokens } Y, \\ k &= 25,000. \end{aligned}$$

191 At time t we have

$$\begin{aligned} P_X(t) &= 4 \text{ USD}, \\ P_Y(t) &= 2 \text{ USD}. \end{aligned}$$

192 Performing the corresponding computations yields:

$$\begin{aligned} x(t) &= \sqrt{25,000 \times \frac{2}{4}} = 111.8034, \\ y(t) &= \sqrt{25,000 \times \frac{4}{2}} = 223.6068, \\ \text{Liquidity}(t) &= 111.8034 \times 4 + 223.6068 \times 2 = 894.4272, \\ \text{Hold}(t) &= 100 \times 4 + 500 \times 1 = 900, \\ \text{LossUsd}(t) &= 900 - 894.4272 = 5.5728, \\ \text{LossPct}(t) &= \frac{5.5728}{900} = 0.619\%. \end{aligned}$$

193 Therefore, there is an impermanent loss of 0.619% due to the price change.

194 3.3 Under a price change, the impermanent loss is invariant with respect 195 to the initial liquidity

196 In the case of a price move, the impermanent loss in percentage terms is constant with respect to the
197 initial liquidity provision. We can see that this is true in Figure 2, where all curves are overlapped,
198 but let us explain the rationale behind.

199 If we increase M times the initial notional of token X , that is $N_X(0)$, then by Rule (2) we
200 need to increase $N_Y(0)$ M times as well. Consequently, the initial number of tokens $x(0)$ and
201 $y(0)$ is increased M times, which in turn means the constant k in Rule (1) is increased M^2 times.
202 When computing all variables at time t all formulas involve M linearly, hence they are all increased
203 M times, in particular $Liquidity(t)$, $Hold(t)$ and $LossUsd(t)$. Finally, in Equation (5) the ratio
204 $LossUsd(t)/Hold(t)$ cancels out the effect of the M factor, and in consequence the impermanent
205 loss in percentage terms is invariant with respect to M .

206 Finally, in Figure 2 we can also observe that whenever there is a price move there is always
207 an impermanent loss, which increases as prices drift away from their initial value, and its rate of
208 increase is faster than linear.

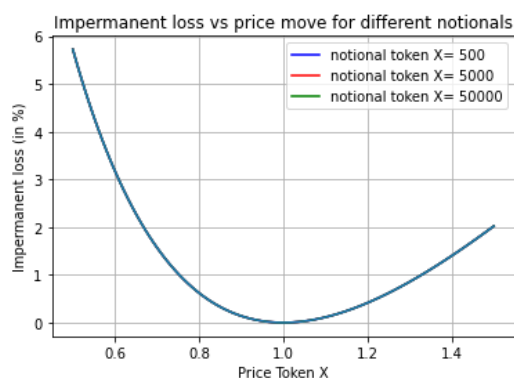


Figure 2: (a) The impermanent loss is always positive on a price move and increases as prices move further away from their initial value. (b) The impermanent loss is constant regardless of the size of our liquidity provision.

209 4 Impermanent loss and liquidity consumption

210 Let us consider the same parameters as in the previous section. The initial conditions are thus:

$$\begin{aligned} P_X(0) &= 5 \text{ USD}, \\ P_Y(0) &= 2 \text{ USD}, \\ N_X(0) &= 500 \text{ USD}, \\ N_Y(0) &= 500 \text{ USD}, \\ x(0) &= 100 \text{ tokens } X, \\ y(0) &= 250 \text{ tokens } Y, \\ k &= 25,000. \end{aligned}$$

211 4.1 Trader buys 5 tokens X from the AMM

212 Let us suppose that there is a trader who wants to buy 5 tokens X with a USD value of 5 each from
213 the AMM: that means a total value of 25 USD. Since the transactions are performed via a swap,
214 the trader will in theory pay 12.5 tokens Y with a USD value of 2 each to settle the trade: that is
215 a total cost of 25 USD. However, due to the rules governing the AMM there will be some slippage
216 i.e. an extra cost in terms of token Y for the trader to close the swap. Let us see how this cost arises.

217

218 The trader buys 5 tokens X at time $t > 0$, hence

$$x(t) = 95 \text{ tokens } X.$$

219 From Rule (1) we can compute the amount of tokens Y that the AMM needs to hold:

$$y(t) = \frac{k}{x(t)} = \frac{25,000}{95} = 263.1579 \text{ tokens } Y.$$

220 From the trader's perspective, the cost of the trade is slightly higher than expected: instead of
221 the theoretical cost of 12.5 tokens Y , the trader pays to the AMM

$$263.1579 - 250 = 13.1579 \text{ tokens } Y$$

222 to settle the swap. This corresponds to a slippage (i.e. extra cost) of

$$13.1579 - 12.5 = 0.6579 \text{ tokens } Y,$$

223 which translates into a slippage of

$$0.6579 \times 2 = 1.3158 \text{ USD},$$

224 or

$$1.3158/25 = 5.263\%.$$

225 From the liquidity provider's perspective, there is a profit on the swap. Indeed, their liquidity
226 and hold positions are

$$\begin{aligned} \text{Liquidity}(t) &= 95 \times 5 + 263.1579 \times 2 = 1,001.3158, \\ \text{Hold}(t) &= 100 \times 5 + 250 \times 2 = 1,000, \end{aligned}$$

227 which implies that the impermanent loss is

$$1,000 - 1,001.3158 = -1.3158 \text{ USD},$$

228 or

$$-1.3158/1,000 = -1.3158\%.$$

229 Since the impermanent loss in this case is negative, the liquidity provider actually made a profit.
230 Let us say there is an "impermanent gain" when the impermanent loss is negative. As before, the
231 impermanent gain will only be realised if we redeem our LP tokens.

232 Notice that the slippage in USD for the trader and the impermanent gain in USD for the liquidity
233 provider is the same. This is true because the swap is a zero sum gain: the frictions incurred by
234 the trader (as slippage) are captured by the liquidity provider (as impermanent gain). In the light
235 of this, we can use the slippage as a proxy for both market impact and liquidity depth within the
236 AMM.

237 4.2 Trader sells 5 tokens X to the AMM

238 In this case the trader who wants to sell 5 tokens X (with a USD value of 5 each) to the AMM and
239 use the proceeds to hopefully buy 12.5 tokens Y (with a USD value of 2 each). However, due to the
240 rules governing the AMM there will be some slippage i.e. an extra cost in terms of token X for the
241 trader to close the swap.

242

243 Since the trader sells 5 tokens X then

$$x(t) = 105 \text{ tokens } X.$$

244 From Rule (1) we necessarily have

$$y(t) = \frac{k}{x(t)} = \frac{25,000}{105} = 238.0952 \text{ tokens } Y.$$

245 From the trader's perspective, the proceeds of the sale of 5 tokens X is not the theoretical 12.5

246 tokens Y but a slightly lower amount:

$$250 - 238.0952 = 11.9048 \text{ tokens } Y.$$

247 In consequence, there is a slippage (i.e. extra cost) of

$$(12.5 - 11.9048) \times 2 = 1.1904 \text{ USD},$$

248 or

$$1.1904/25 = 4.762\%.$$

249 From the liquidity provider's perspective:

$$Liquidity(t) = 105 \times 5 + 238.10 \times 2 = 1,001.1904,$$

$$Hold(t) = 100 \times 5 + 500 \times 1 = 1,000,$$

250 and therefore their impermanent gain in the swap is

$$1,001.1904 - 1,000 = 1.1904$$

251 or

$$1.1904/1,000 = 0.119\%.$$

252 Again, the slippage and the impermanent gain in USD are equal.

253 **4.3 Impermanent gain and slippage are decreasing in the available liq-** 254 **uidity**

255 As we can see in Figure 3, if the available liquidity in the AMM increases then eventually both the
256 impermanent gain and the slippage reduce to zero. Therefore, the exchange becomes more efficient
257 with more available liquidity, since the friction or market impact (either impermanent loss/gain or
258 slippage) disappears. Moreover, in USD terms the slippage of the trader and the impermanent gain
259 of the liquidity provider are equal: in a zero-sum game "one (wo)man's fun is another's hell".

260 **5 Slippage and liquidity balance/imbalance within the AMM**

261 Even if prices are the same, if the available liquidity within the AMM is not balanced (i.e. 50/50 in
262 notional), the swap rate offered by the AMM will change. Let us see how this is true.

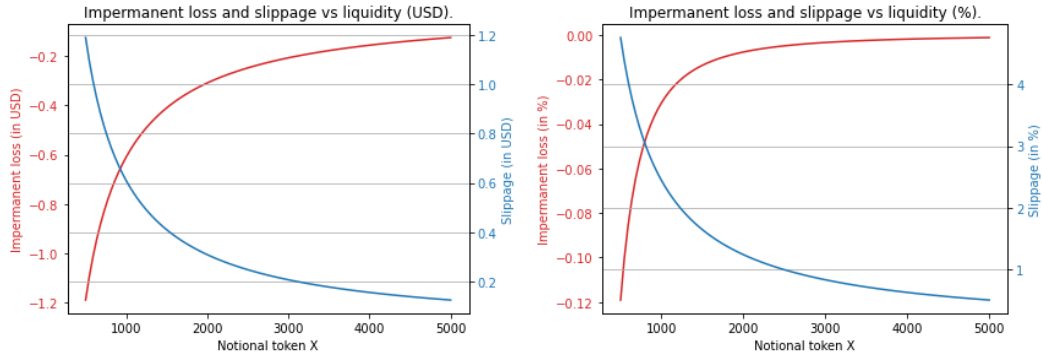


Figure 3: When there is a trade (a) the negative impermanent loss of the liquidity provider (hence a profit) is increasing with respect to the AMM volume in USD, and (b) the slippage of the trader is decreasing. Eventually, both converge to zero.

5.1 Trader buys in a balanced liquidity pool

Assume a 50/50 liquidity pool i.e. a balanced pool. As usual, we assume the following values:

$$\begin{aligned} P_X(0) &= 5 \text{ USD}, \\ P_Y(0) &= 2 \text{ USD}, \\ N_X(0) &= 500 \text{ USD}, \\ N_Y(0) &= 500 \text{ USD}. \end{aligned}$$

Therefore,

$$\begin{aligned} x(0) &= 100 \text{ tokens } X, \\ y(0) &= 250 \text{ tokens } Y, \\ k &= 100 \times 250 = 25,000. \end{aligned}$$

If the trader buys 5 tokens X then

$$\begin{aligned} x(t) &= 100 - 5 = 95, \\ y(t) &= \frac{k}{x(t)} = \frac{25,000}{95} = 263.1579. \end{aligned}$$

In the end, the price the trader pays for the swap is

$$\text{TraderCost}(t) = y(t) - y(0) = 263.1579 - 250 = 13.1579 \text{ tokens } Y.$$

268 5.2 Trader buys the token with smaller liquidity in the pool

269 Let us assume that the liquidity pool is not balanced and that there are more tokens Y than tokens
270 X in notional. More precisely, suppose that we have

$$\begin{aligned}P_X(0) &= 5 \text{ USD}, \\P_Y(0) &= 2 \text{ USD}, \\x(0) &= 50 \text{ tokens } X, \\y(0) &= 500 \text{ tokens } Y.\end{aligned}$$

271 Therefore,

$$\begin{aligned}k &= 50 \times 500 = 25,000, \\N_X(0) &= 50 \times 5 = 250 \text{ USD}, \\N_Y(0) &= 500 \times 2 = 1000 \text{ USD}.\end{aligned}$$

272 If the trader buys 5 tokens X then

$$\begin{aligned}x(t) &= 50 - 5 = 45, \\y(t) &= \frac{k}{x(t)} = \frac{25,000}{45} = 555.5556, \\TraderCost(t) &= y(t) - y(0) = 555.5556 - 500 = 55.5556 \text{ tokens } Y.\end{aligned}$$

273 Since direction of the trader's swap is increasing the imbalance, they get a worst swap rate than
274 in the balanced case. Indeed, given the big imbalance in notionals and the constant-product formula,
275 the extra cost is ridiculously high:

$$\frac{55.5556}{13.1579} - 1 = 322.22\%$$

276 more expensive than the balanced case.

277 5.3 Trader buys the token with greater liquidity in the pool

278 Let us assume that there are more tokens X than tokens Y in notional. More precisely, suppose
279 that we have

$$\begin{aligned}P_X(0) &= 5 \text{ USD}, \\P_Y(0) &= 2 \text{ USD}, \\x(0) &= 200 \text{ tokens } X, \\y(0) &= 125 \text{ tokens } Y.\end{aligned}$$

280 Therefore,

$$\begin{aligned} k &= 200 \times 125 = 25,000, \\ N_X(0) &= 200 \times 5 = 1000 \text{ USD}, \\ N_Y(0) &= 125 \times 2 = 250 \text{ USD}. \end{aligned}$$

281 If the trader buys 5 tokens X then

$$\begin{aligned} x(t) &= 200 - 5 = 195, \\ y(t) &= \frac{k}{x(t)} = \frac{25,000}{195} = 128.2051, \\ \text{TraderCost}(t) &= y(t) - y(0) = 128.2051 - 125 = 3.2051 \text{ tokens } Y. \end{aligned}$$

282 Since the direction of the trader's swap is helping to reduce the imbalance, they get a better
283 swap rate than in the balanced case, resulting in an incredibly large price discount:

$$\frac{3.2051}{13.1579} - 1 = -75.641\%.$$

284 These examples are of course very dramatic in liquidity imbalance and unlikely to happen in
285 reality. However, they highlight how important it is for liquidity providers and swap traders to
286 deal with balanced pools when the AMMs follow the constant-product formula. In addition, the
287 examples stress the potential profitability of arbitrages when the pools are not balanced.

288 6 Impermanent loss and price reversion

289 6.1 The impermanent loss disappears when there is price reversion

290 Let us assume the usual initial conditions as before, i.e.

$$\begin{aligned} P_X(0) &= 5 \text{ USD}, \\ P_Y(0) &= 2 \text{ USD}, \\ N_X(0) &= 500 \text{ USD}, \\ N_Y(0) &= 500 \text{ USD}, \\ x(0) &= 100 \text{ tokens}, \\ y(0) &= 250 \text{ tokens}, \\ k &= 25,000. \end{aligned}$$

291 The initial liquidity provision is

$$\text{Liquidity}(0) = N_X(0) + N_Y(0) = 500 + 500 = 1,000 \text{ USD}. \quad (12)$$

292 Suppose that at $t = t_1$ the price of token X drops: $5 \rightarrow 4$ USD:

$$\begin{aligned} P_X(t_1) &= 4 \text{ USD}, \\ P_Y(t_1) &= 2 \text{ USD}. \end{aligned}$$

293 Therefore, as before, we have

$$\begin{aligned} x(t_1) &= \sqrt{k \frac{P_Y(t_1)}{P_X(t_1)}} = \sqrt{25,000 \times \frac{2}{4}} = 111.8034, \\ y(t_1) &= \sqrt{k \frac{P_X(t_1)}{P_Y(t_1)}} = \sqrt{25,000 \times \frac{4}{2}} = 223.6068, \\ N_X(t_1) &= x(t_1)P_X(t_1) = 111.8034 \times 4 = 447.2136, \\ N_Y(t_1) &= y(t_1)P_Y(t_1) = 223.6068 \times 2 = 447.2136. \end{aligned}$$

294 The liquidity provision at $t = t_1$ is thus

$$Liquidity(t_1) = N_X(t_1) + N_Y(t_1) = 447.2136 + 447.2136 = 894.4272, \quad (13)$$

295 which represents a loss of $1,000 - 894.4272 = 105.5728$ USD with respect to the initial liquidity
296 supply that the liquidity provider deposited within the AMM.

297 In $t = t_2$ the price of token X returns to its initial value: $4 \rightarrow 5$ USD:

$$\begin{aligned} P_X(t_2) &= 5 \text{ USD}, \\ P_Y(t_2) &= 2 \text{ USD}. \end{aligned}$$

298 Therefore,

$$\begin{aligned} x(t_2) &= \sqrt{k \frac{P_Y(t_2)}{P_X(t_2)}} = \sqrt{25,000 \times \frac{2}{5}} = 100, \\ y(t_2) &= \sqrt{k \frac{P_X(t_2)}{P_Y(t_2)}} = \sqrt{25,000 \times \frac{5}{2}} = 250, \\ N_X(t_2) &= x(t_2)P_X(t_2) = 100 \times 5 = 500, \\ N_Y(t_2) &= y(t_2)P_Y(t_2) = 250 \times 2 = 500. \end{aligned}$$

299 The liquidity provision at $t = t_2$ is then

$$Liquidity(t_2) = N_X(t_2) + N_Y(t_2) = 500 + 500 = 1,000. \quad (14)$$

300 From Equations (12)-(14) we observe that the impermanent loss due to the price drop of token
301 X at $t = t_1$ was completely reverted when the price came back to its initial value at $t = t_2$. This is
302 exactly why it is called an “impermanent” loss: it reverts to zero when prices return to their initial

303 state.

304 6.2 The impermanent loss may become permanent if some liquidity is 305 withdrawn

306 As we saw in the previous section, the impermanent loss reverted to zero when the prices reverted
307 to their initial state at $t = t_2$. This is a general principle, not a particular feature from our example.
308 Indeed, observe the pricing formulas under a price move in Equation (11):

$$\begin{aligned}x(t_2) &= \sqrt{k \frac{P_Y(t_2)}{P_X(t_2)}}, \\y(t_2) &= \sqrt{k \frac{P_X(t_2)}{P_Y(t_2)}}.\end{aligned}$$

309 Not only these expressions have no dependence of the previous states but they only depend on the
310 current price levels $P_X(t_2)$ and $P_Y(t_2)$ as well as the constant k . Under the assumption of prices
311 reverting i.e. $P_X(t_2) = P_X(0)$ and $P_Y(t_2) = P_Y(0)$, the only way we can have $x(t_2) \neq x(0)$ and/or
312 $y(t_2) \neq y(0)$ is by varying k . In normal conditions this is impossible because, by definition of the
313 constant-product formula, k is constant. However, if before the price reversion is completed we
314 withdraw some liquidity then effectively k can decrease, and in consequence the final liquidity may
315 differ from the initial liquidity.

316
317 Let us see this more clearly in an example. Let us assume the chain of events in the previous
318 section and place ourselves at $t = t_1$:

$$\begin{aligned}P_X(t_1) &= 4, \\P_Y(t_1) &= 2, \\x(t_1) &= 111.8034, \\y(t_1) &= 223.6068, \\N_X(t_1) &= 447.2136, \\N_Y(t_1) &= 447.2136, \\Liquidity(t_1) &= 894.4272.\end{aligned}$$

319 Imagine now that at time $t' \in (t_1, t_2)$ we panicked. More precisely, suppose that we decided to
320 remove 10% of our liquidity provision from the AMM and exchange it to USD. This means that we
321 have

$$PanicCash(t') = Liquidity(t_1) \times 0.1 = 894.4272 \times 0.1 = 89.4427 \text{ USD}$$

322 in addition to our remaining liquidity provision. In consequence, we have the following state at

323 $t = t'$:

$$\begin{aligned}P_X(t') &= 4, \\P_Y(t') &= 2, \\x(t') &= 111.8034 \times 0.9 = 100.6231 \\y(t') &= 223.6068 \times 0.9 = 201.2461.\end{aligned}$$

324 Since the liquidity provision within the AMM has changed, the constant k needs to be recom-
325 puted:

$$k = x(t')y(t') = 100.6231 \times 201.2461 = 20,250 < 25,000.$$

326 Now let us place ourselves at time $t = t_2$ where the price reversion occurred:

$$\begin{aligned}P_X(t_2) &= 5, \\P_Y(t_2) &= 2, \\k &= 20,250.\end{aligned}$$

327 In consequence,

$$\begin{aligned}x(t_2) &= \sqrt{k \frac{P_Y(t_2)}{P_X(t_2)}} = \sqrt{20,250 \times \frac{2}{5}} = 90, \\y(t_2) &= \sqrt{k \frac{P_X(t_2)}{P_Y(t_2)}} = \sqrt{20,250 \times \frac{5}{2}} = 225, \\Liquidity(t_2) &= x(t_2)P_X(t_2) + y(t_2)P_Y(t_2) \\&= 90 \times 5 + 225 \times 2 = 900.\end{aligned}$$

328 The total value of our portfolio is thus

$$Portfolio(t_2) = PanicCash(t_2) + Liquidity(t_2) = 89.4427 + 900 = 989.4427,$$

329 which is strictly smaller than the initial portfolio

$$Portfolio(0) = Liquidity(0) = 1,000.$$

330 7 Conclusions

- 331 1. AMMs are close relatives of MMs: if we see the security as a first token and the currency paid
332 for said security as a second token, dealing with an MM becomes a swap in the AMM sense.
- 333 2. The slippage a trader incurs when dealing with AMMs can be interpreted as the half spread
334 in MMs: both are deviations from the current market price due to the trade. As such, they
335 can be considered as a proxy for market impact and depth of liquidity within an AMM.

- 336 3. Under the effect of a price move, the impermanent loss in percentage is invariant to the available
337 liquidity: no matter how big is the available liquidity, AMMs will always lose the same amount
338 in percentage points when prices move. This is very similar to the “adverse selection” that
339 traditional MMs suffer when providing liquidity.
- 340 4. The impermanent loss due to price moves grows faster than linear.
- 341 5. AMMs become more efficient as their liquidity increases: the slippage decrease and converges
342 to zero.
- 343 6. Having a balanced pool in notional terms is crucial for price discovery and to maintain the
344 current price levels. If the pool is imbalanced then, compared to the balanced case, the swap
345 cost for the trader will be higher (resp. lower) than expected if their transaction amplify (resp.
346 decrease) the current imbalance.
- 347 7. The price differences due to liquidity imbalances can be quite dramatic: due to the constant-
348 product formula, the prices move away from their equilibrium following a power law dynamics,
349 which is much faster than linear.
- 350 8. The impermanent loss disappears when there is price reversion: all state variables return to
351 their initial values, hence it is as if “nothing had happened”.
- 352 9. However, if we withdraw some liquidity before the price reversion is completed, we could realise
353 some of the impermanent loss, effectively rendering it “permanent”.

354 8 See also

355 A good entry point to the world of Decentralised Finance (DeFi) and cryptocurrencies are the works
356 of Campbell R. Harvey. Mr. Harvey wrote a paper [HRS21a], which later became a book [HRS21b],
357 and whose content is covered in detail in his Coursera Specialisation in DeFi [Har].

358 The most important AMM in the DeFi space is Uniswap in the Ethereum blockchain [Unib]: al-
359 though Uniswap is second-place in total value locked (TVL, equivalent of market share) in Ethereum
360 behind Curve [Cur] (according to DeFi Llama [Llaa]), it was the first DeFi offering AMMs and swaps
361 in any blockchain. In the Uniswap Docs page there is extensive information about AMMs, including
362 the constant-product formula [Unia], whilst in the Research section [Unic] there are some academic
363 papers on AMMs. There are other several AMMs in the Ethereum blockchain that have a high rank
364 of TVL e.g. Sushiswap [Sus]. For the Solana blockchain, Raydium [Rayb] and Saber [Sab] are the
365 biggest AMMs in TVL (according to DeFi Llama [Llab]), although there are a few other AMMs
366 that are increasing their size and are worth to mention e.g. Orca [Orc].

367 There are other online resources that are useful. For a quick review of AMMs, liquidity pools,
368 staking and yield farming we suggest to start with the blog [Aca]. For a simple calculator of
369 impermanent loss under a price move you could visit [dai].

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